HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Applied fundamental engineering principles to support the design and performance of air-cooled chiller components for the European market.
- Collected, reduced, and presented test data from laboratory programs under engineer guidance, contributing to product development.
- Assisted in the selection of mechanical components, including heat exchangers and fans, evaluating them based on cost and performance requirements.
- Gained understanding of European mechanical and safety codes, including CE marking and the Pressure Equipment Directive (PED), to ensure product compliance.
- Supported factory operations and product technical service by providing day-to-day assistance on assigned chiller components.
- Participated in design reviews and contributed to engineering deliverables, including design specifications, under close supervision.
- Developed Python scripts to automate data collection and reporting for laboratory testing, reducing manual processing time by approximately 30%.

SKILLS

Core Engineering – Mechanical Design, Fluid Dynamics, Thermal Principles, Refrigeration Systems, Component Selection

Standards & Compliance – CE Marking, Pressure Equipment Directive (PED), Machinery Directive, EN 378, EN 14511, EU F-Gas Regulation, Ecodesign / ErP framework

Data Analysis & Tools – Data Reduction, Data Presentation, Statistical Analysis, Power BI, Tableau, R / RStudio

Software Development – Python, Java, Spring Boot, SQL, REST APIs, Docker, Linux

Cloud & CI/CD – AWS (EC2, S3, Lambda, CloudWatch), Jenkins, Git

Project Management & Support – Agile Methodologies, Jira, Confluence, Change Management, Technical Support, Laboratory Testing, Factory Operations

PROJECTS

Automated Chiller Performance Data Analysis

- Developed Python scripts to ingest and process raw data from laboratory chiller tests, automating the reduction and aggregation of performance metrics.
- Created **Power BI dashboards** to visualize key performance indicators (KPIs) such as thermal efficiency and component load, enabling engineers to quickly identify trends.
- Implemented data validation checks to ensure accuracy of collected test data, improving data integrity for subsequent analysis and reporting.

Component Selection & Integration Support Tool

- Designed and built a Java Spring Boot application to assist in the selection and comparison of mechanical components (fans, heat exchangers) based on specified performance criteria and cost constraints.
- Integrated the application with a **PostgreSQL** database to manage component specifications and historical performance data.
- Developed a RESTful API to allow other internal systems to query component availability and performance characteristics.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)